React Redux

🔷 What is Redux (in React)?

Redux is a **state management tool** used with React to manage and share data across components.

🔥 Why we use Redux?

1. 📦 Centralized State (Single Source of Truth)

Instead of passing data from parent → child → child (props drilling), Redux stores everything in **one global store**.

👉 Any component can access data directly.

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2. 🔄 Easy State Sharing

If multiple components need the same data (like user info, cart, theme):

❌ Without Redux → pass props manually
✅ With Redux → access from store anywhere

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3. 🧠 Predictable State Management

Redux uses a clear flow:

Action → Reducer → Store → UI update

This makes your app behavior **easy to understand and debug**.

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4. 🛠️ Better for Large Applications

For small apps → Redux not needed
For big apps → Redux helps manage complex data easily

Examples:

- E-commerce (cart, orders)

- Social media apps
- Dashboards

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5. 🐞 Easy Debugging

With tools like **Redux DevTools**, you can:

- Track state changes
- Undo/redo actions
- Debug easily

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6. ⚡ Performance Optimization

Redux helps avoid unnecessary re-renders by updating only required components.

🧩 Simple Example

Without Redux:

App → Component A → Component B → Component C
(pass props step by step)

With Redux:

Store (global)
↙ ↓ ↘
A B C

🧠 When to use Redux?

Use Redux if:
✓ Many components share data
✓ App state is complex
✓ You need global state (auth, cart, etc.)

Avoid Redux if:
❌ Small app
❌ Simple state (use React useState / useContext instead)

⚡ Bonus (Modern Way)

Today, we use **Redux Toolkit (RTK)** instead of plain Redux
👉 Less code + easier setup

🔚 Final Line

👉 Redux is used to manage and share global state in a clean, scalable, and predictable way in React apps.

🚀 1. Create Vite React App

```
npm create vite@latest redux-vite-app
cd redux-vite-app
npm install
npm install @reduxjs/toolkit react-redux
npm run dev
```

👉 Choose: **React + JavaScript**

📂 2. Folder Structure (.jsx)

```
src/
 ├── app/
 |   └── store.jsx
 ├── features/
 |   └── counter/
 |       ├── counterSlice.jsx
 |       └── Counter.jsx
 ├── App.jsx
 └── main.jsx
```

⚙️ 3. Store Setup

📄 src/app/store.jsx

```
import { configureStore } from '@reduxjs/toolkit';
import counterReducer from '../features/counter/counterSlice';

export const store = configureStore({
  reducer: {
    counter: counterReducer,
  },
});
```

🔢 4. Counter Slice

📄 src/features/counter/counterSlice.jsx

```
import { createSlice } from '@reduxjs/toolkit';

const initialState = {
  value: 0,
};

const counterSlice = createSlice({
  name: 'counter',
  initialState,
  reducers: {
    increment: (state) => {
      state.value += 1;
    },
    decrement: (state) => {
      state.value -= 1;
    },
  },
});

export const { increment, decrement } = counterSlice.actions;
export default counterSlice.reducer;
```

🧩 5. Counter Component

📄 **src/features/counter/Counter.jsx**

```
import { useSelector, useDispatch } from 'react-redux';
import { increment, decrement } from './counterSlice';

const Counter = () => {
  const count = useSelector((state) => state.counter.value);
  const dispatch = useDispatch();

  return (
    <div>
      <h2>Count: {count}</h2>
      <button onClick={() => dispatch(increment())}>+</button>
      <button onClick={() => dispatch(decrement())}>-</button>
    </div>
  );
};

export default Counter;
```

🔗 6. Connect Redux to Vite App

📄 **src/main.jsx**

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App.jsx';
import { Provider } from 'react-redux';
import { store } from './app/store.jsx';

ReactDOM.createRoot(document.getElementById('root')).render(
  <React.StrictMode>
    <Provider store={store}>
      <App />
    </Provider>
  </React.StrictMode>
);
```

🏠 7. App Component

📄 **src/App.jsx**

```
import Counter from './features/counter/Counter';

function App() {
  return (
    <div>
      <h1>Redux + Vite (.jsx)</h1>
      <Counter />
    </div>
  );
}

export default App;
```

▶️ **8. Run Project**

```
npm run dev
```

🔷 **1. Import statements**

import { useSelector, useDispatch } from 'react-redux';

useSelector 👉 used to read data from Redux store

useDispatch 👉 used to send actions to Redux store

```
import { increment, decrement } from './counterSlice';
```

Importing actions from your Redux slice

increment() 👉 increase value

decrement() 👉 decrease value